Reliable Team Assistant: Phase 1

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Question and implemented scope

Can dependency-aware recovery preserve evidence-backed diagnosis quality with less recomputation after a trusted invalidation event? Configuration and telemetry roles collect read-only snapshots; a coordinator diagnoses. The runtime records consumed versions, invalidates dependent work, retains independent artifacts, rejects obsolete publications and links corrections. Missing recorded provenance triggers restart in development controls; this fallback was not exercised in the formal batch. Arbitrary error discovery, operational writes and invisible dependencies are outside the implemented guarantee.

Workload and frozen protocol

Cloud-OpsBench is pinned to ea05dafbc87b5985df8721622d63d4229c2b4224. Formal cases boutique/runtime/32, service/12 and performance/5 each run twice, reversing restart/selective order on repeat 2. These are three independent cases, not six. TritonAI api-glm-5.3 uses low reasoning, temperature 0 and 8192 output tokens. Course DataHub CPU 2/8 GB runs isolated Python 3.11.9 with exact dependency pins. Source 1380c6b identifies execution; the complete freeze is archived.

The user advanced launch to October 24 before formal observations were inspected. Cases, order, model, budgets and scoring were unchanged. Course acceptance of the explicitly user-selected benchmark remains unverified. Earlier development experiments and the 17-call TritonAI seen-case workflow remain separate feasibility evidence. No email or assignment submission was sent.

Formal quality results

All 6 units completed. Strict pass counts over all 6 assigned units: clean 4/6; restart 3/6; selective 3/6; prevention 3/6; expiry-restart 2/6; expiry-selective 2/6; no-recovery 0/6. For numeric-port recovery, only 4 units were eligible: restart and selective each passed 3/4, with different failed units. The other 2 units remain not-injected, not quietly excluded. Collection is not a diagnosis endpoint. No-recovery fails the current-publication criterion by construction; 0/6 does not imply six incorrect diagnoses. Prevention checks source integrity, not arbitrary semantic truth.

Strict success requires diagnosed status, native joint root/target accuracy and evidence closure, input-supported reasoning, current publication, valid output contract and budget. Input-only judgments on all 36 produced answers were hash-sealed before native labels were accessed. Codex performed the review; it is not independent human validation. All 54 assigned arm slots remain in scoring. Provisional publication does not mean operational approval.

Development, costs and remaining evidence

Formal measurement boundary

Four eligible formal units: selective used 1 versus 3 restart calls and 33,260 versus 64,902 incremental tokens (48.8% fewer). Both passed 3/4 with different failures; quality equivalence is not established. None of four corrupted ports propagated. Both used 3 calls on all six all-expired units. Formal timing sums API latency; full workflow latency and tracking overhead were not recorded.

Handoff and temporal generation development

Earlier runtime/1 handoff: three repeats, 54 calls, $0.21365440; R/S used 6/3 calls with 8 retained artifacts. All 15 core diagnoses were source-supported but no false claim propagated; ancillary timing errors remained. New calibration used 6 original/revised calls, then 3 JSON-mode calls, then 3 generic evidence-contract calls on fixed seen inputs. Revised v3 had 1/3 malformed JSON; v4/v5 each 3/3 valid. In v5, all 3 primary diagnoses/actions stayed supported and observed cadence/count confusions were absent, but health, interpretation and coverage overclaims remained. These adaptive, known-error samples do not establish generalization.

Cross-case eligibility and input-cap failure

Service/2 two assigned repeats consumed 11 calls. Repeat 1 joint liveness/readiness wording had no unique safe mutation span; fault/recovery arms remained not-injected. Repeat 2 clean retrieval made a 40,068-byte request plus 512 reserve, 580 over the 40,000 cap; failure occurred before another API call. No eligible fault exposure or recovery pair resulted. An earlier missing-snapshot startup used 0 calls and is also retained. The one completed clean answer supported a DNS lookup failure mechanism but overclaimed other facts.

Explicit seeded conditional recovery

One saved runtime/1 state was seeded with a wrong protocol and no-change recommendation, then invalidated. R/S used 6/3 calls, 31,142/15,638 tokens (49.8% fewer); selective retained 8 valid artifacts and both superseded obsolete publication. Input-only AI review supported both final diagnoses and corrective actions; minor ancillary wording/coverage issues remained. Measured workflow time 35.54/15.25 s includes invalidation, cloning, calls, validation, publication and persistence, excluding fixed restore, notification delivery and clock write. One fixed-order pair is not a latency estimate. Wrong descendants were artificially seeded and not reread: this tests replacement after trusted invalidation, not natural propagation or self-discovery.

Costs and decision

Frozen formal remains 92 calls, 519,696 tokens, $0.40225095. This continuation: 32 calls, 175,193 tokens, $0.18411535, reconciled against account delta. All development receipts total $0.66344450 of $5; course balance $48.92789740. No API retries or personal billing. Next work is byte-aware retrieval and independently scoped configuration facts, with offline gates before a new versioned pilot. Full-answer equivalence and cross-case recovery benefits remain unproven.

Reproduction and timeline appendix

Immutable evidence and new development records

The package preserves formal results, review seal, 36 answers and 54 slots byte-for-byte. The nested historical Phase1-Evidence.zip also preserves original execution source; the current provider has opt-in development extensions and deliberately fails the old execution freeze. Added development records include all 32 calls, zero-call startup, unrun/ineligible assignments, review seals, accounting and offline input-cap reconstruction.

Offline verification checklist

Use Python >=3.11 and requirements.lock. Verify the original formal release, summarize saved handoff evidence, and replay prior timing/synthetic controls. For each new batch, validate input/config/contract hashes, slot counts, request pairing, unchanged model settings, JSON-format option and instruction changes; sum receipt costs separately from conservative budget charges. Check independent-review seals and retain failed outputs. Exact inputs and acceptance checks are listed in docs/phase1-autonomous-reproduction-checklist-20261024.txt. Saved-output validation does not regenerate answers or independently prove semantics.

Completed October 24

Formal evaluation and three-repeat seen-case handoff, offline controls, then three versioned live calibration stages. Prompt refinements used observed development errors. Independent AI review found useful targeted improvement with unresolved ancillary claims. AI assistance includes implementation, experiments, analysis and report drafting; neither AI review nor student review should be called independent human adjudication.

Remaining course and interpretation checks

Confirm student review, course benchmark acceptance and exact live deadline/upload slot. Saved overview lists November 4 for Phase 1; acceptance and deadline hour remain unverified. Neither source-supported recommendations nor version-valid publication means operational remediation was executed. DataHub was stopped after evidence download. No submission or email was sent.

Remaining term

Preserve source verification even when it prevents organic propagation. Conditional seeded recovery measures correction given a wrong state, not natural error frequency. Broader evaluation requires a genuinely unexposed set, leakage audit and new frozen protocol. Multi-group scheduling and gated learning follow core recovery evidence. Prepare final report/presentation after checking official dates. Static snapshots, trusted invalidation, mutable model aliases and small development samples limit claims. No email or assignment submission has been sent.